

Capstone Project Proposal

Name: [REDACTED]

Objective

[Describe the overall objective of the project. If necessary, explain the context and motivation.]

As a Data Science and Japanese double major, I want to connect these two vastly different disciplines in this capstone project. Coming into this class, I knew I wanted to focus on population shrinkage in Japan and try to find some correlation between independent variables that may help us create a predictive model to better understand population changes in the future. Population shrinkage and aging is one of Japan's largest issues today, and many cities do not have the necessary infrastructure to provide for a majority older population, so I thought this would be an interesting topic to explore further.

As I narrowed down my topic, as I started out with a much broader and much less clear idea, I noticed that a great deal of research has been put into areas surrounding mainland cities like Tokyo, Kyoto, and Osaka, but very little efforts have been put towards understanding this issue in Hokkaido, Japan's northern-most and second largest island. Therefore, I would like to focus my project on the population shrinkage in Sapporo, Hokkaido's largest city, and its suburbs. The overall objective of this project would be to analyze and model patterns of population shrinkage with the goal of finding key features that correlate with the negative population rates. I will likely focus on access to public transportation as my independent factor.

Outcome

[Clearly state the final outcome of the capstone, e.g., predictive model, statistical analysis, dashboard, UI, etc]

By the end of this semester, I would like to have completed all of my statistical analysis and a rough draft of my paper. Next semester, not for course-credit, I would like to finalize and publish my paper if possible.

Impact

[In a few sentences, delineate the potential impact of the project. Who would benefit from your deliverables?]

I would target this project towards other researchers/academic scholars. It could also be useful to local city governments and community organizations focused on countering depopulation in Japan. It would certainly be most useful to those focused on Hokkaido, but when combined with

studies done on other major areas it could help create a more comprehensive understanding of the nation-wide crisis.

Dataset(s)

[List your data sources with links to them. If you have already uploaded them to your capstone repository on GitHub, please mention the location. In addition, briefly discuss the datatypes and the reliability of the data.]

- Population Data:
 - o [Population Census 2020 Population Census Divided Version of Statistical Table 01:Hokkaido | File | Browse Statistics | 政府統計の総合窓口](#)
 - The Population Census of Japan is conducted every 5 years, going back to 1920. Hokkaido itself in this census is broken down into cities, and then those are further broken down into wards. The datatypes in this dataset are entirely numerical. These datasets are from the Japanese government, and to my knowledge, are the most reliable data available. I have links to all of census years, and will be using likely a 20-35 year range, but I have only included one census here for length's sake.
- Transportation data:
 - o [GTFS Transit Data for Japan - gtfs-sapporo1 by Hokkaido-open-data - busmaps.com](#)
 - o [Public Transport GTFS Data - Akan Bus - HODA](#)
 - This data comes from Hokkaido-open-data and includes the Sapporo bus routes. There is a larger dataset that, once broken down, should provide the same information for surrounding cities. This isn't provided from the government, but it does seem to be a reputable and trusted source for transit data. I know that the one from busmaps.com does not seem as trustable, and I will certainly look more into it before using it, but at a first glance it really does just seem to be a translation of the original source, which is in Japanese, with some slight reformatting.
 - o I haven't found a dataset for the train systems yet, but I have found lists of stops and locations, along with the metro map itself. I can always input the data into a spreadsheet myself before using it for analysis. I will certainly be looking into other datasets as the semester progresses, but this is just what I have found as of now.

Approach

[Talk about how you plan on approaching this capstone through several steps. List the steps below.]

Step 1: Find relevant papers from similar studies discussing suburban population shrinkage outside of Japan's major cities. Find datasets containing population rates of Japan over the last twenty years. Find education, transportation, and healthcare access datasets for Japan as well. It

would be useful if I could find data including age distribution, birth rates, and migration rates as well.

Step 2: Narrow down my datasets to focus on the Hokkaido region.

Step 3: Run data analysis and create models. In the end, I would like to determine correlation, potential causation, and predict future population shrinkage in Sapporo and its suburbs.

Step 4: Data visualization

Step 5: Conclusions, Report (End of semester)

Step 6: Draft paper, reach out to journal for publication

Step 7: Finalize paper with any necessary edits

Timeline

[Edit the following example timeline:

My proposed timeline:

- (3 weeks) Finalize idea, collect datasets, read related papers
- (2 weeks) Cleaning and organizing data, read papers
- (5 weeks) Data Analysis, modeling
- (2 weeks) Analyzing results
- (1 week) Compile results
- (1 week) Draft report
- (1 week) Poster and Final Presentation

Possible Issues

[List some of the prospective challenges and issues and discuss how you envision overcome them]

- Difficulties finding city-level data is certainly harder to find. I am going to try to look at national level data first and try to see how small the categories are broken down. If they don't have city-level data, I will look on smaller scale Japanese government websites.
- Language barriers are certainly a concern. While many of Japan's national datasets have been translated into English, that likely isn't true for the smaller, city-level datasets. I have an intermediate Japanese level, which is not sufficient for academic research. However, I am certainly up to the challenge and think this will be a great opportunity to gain experience with new vocabulary and complex grammar structures that I otherwise wouldn't be exposed to.
- Statistics is certainly not my strong suit, so in the beginning I expect to struggle to figure out which models and analysis are appropriate/necessary to achieve my goals. I will do extra reading and research to figure out which are most appropriate considering my variable types and end goals.
- Code errors are inevitable. When running into small issues with my code, I may prompt AI (ChatGPT is pretty good at finding missing punctuation in my experience). If that doesn't fix the problem, I will try the data science help desk.